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Portable Coordinate Measuring Arm Joint Design

Mechatronic Project 478
Final Report

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Executive summary

Title of Project
Joint Design for a Portable Coordinate Measuring Arm (PCMA)
Objectives
To derive technical performance measures (TPMs) from an error-propagation model, and use them to design, build, and test a position-holding joint that maintains and /or improves the arm's overall accuracy.
What is current practice and what are its limitations?
Current joints rely on counterbalancing and lack holding torque, meaning the operator must physically support the arm's weight during operation.
What is new in this project?
The introduction of a position-holding mechanism specifically engineered using mathematically derived TPMs.
If the project is successful, how will it make a difference?
It will directly maintain or improve the physical measurement accuracy of the arm. Additionally, it will yield a modified kinematic model to serve as a generalized framework for future accuracy analysis.
What are the risks to the project being a success? Why is it expected to be successful?
Risks: University workshop manufacturing capabilities may fall short of required tolerances, and the holding mechanism could introduce unwanted friction. Expectation of Success: The design requirements are heavily grounded in an established, proven error model.
What contributions have/will other students made/make?
The project builds directly upon an initial error-propagation model developed by Stellenbosch alumna Leon Hall.
Which aspects of the project will carry on after completion and why?
The modified error-propagation model. It will be able to accept custom arm dimensions and compute the uncertainty distribution within a given work volume, forming the mathematical foundation for future PCMA development.
What arrangements have been/will be made to expedite continuation?

Extensive, standardized documentation will be left behind to ensure future researchers can build directly onto this work without repeating groundwork.

ECSA Exit Level Outcomes

This section addresses the following Engineering Council of South Africa (ECSA) exit level outcomes and indicates how each outcome was demonstrated in the project.

GA 1 - Problem-Solving

The project addresses an engineering problem, which is the lack of a joint design for PCMA's that meet accuracy requirements that account for spatial dependency and proper position-holding implementation without the impedance of uninhibited movement.

GA 2 - Application Scientific and Engineering Knowledge

GA 3 - Engineering Design

GA 5 - Use of Engineering Tools

GA 6 - Professional and Technical Communication

GA 8 - Individual and Collaborative Teamwork

GA 9 - Independent Learning Ability

Acknowledgements

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List of symbols

Constants

$$L_0 = 300 \text{ mm}$$

Variables

Re_D	Reynolds number (diameter)	[]
x	Coordinate	[m]
$\ddot{x}$	Acceleration	[m/s ²]
θ	Rotation angle	[rad]
τ	Moment	[N·m]

Vectors and Tensors

$\vec{v}$ Physical vector, see equation ...

Subscripts

a	Adiabatic
a	Coordinate

Abbreviations

3D	three-dimensional
AI	artificial intelligence
CAD	computer-aided design
D-H	Denavit-Hartenberg
DIP	design independent parameter
DOF	degrees-of-freedom
EC	evaluation criteria
EEPROM	electrically erasable programmable read-only memory
EMF	electromotive force
FBD	freebody diagram
FEA	finite element analysis

FR	functional requirement
GPIO	general pin input/output
HMI	human machine interface
ISR	interrupt service routine
MOSFET	metal-oxide-semiconductor field-effect transistor
NSoI	narrower system of interest
PCMA	portable coordinate measuring arm
PPR	pulse per revolution
PSD	position sensitive detector
SCADA	supervisory control and data acquisition
SPAT	single-point articulation test
TPM	technical performance measure
WSoI	wider system of interest

Chapter 1

Introduction

1.1 Background

A portable coordinate measuring arm (PCMA) is a handheld metrology instrument that captures three-dimensional (3D) coordinates directly from a physical object, as shown in Figure 1.1 below (1).



Figure 1.1: A portable coordinate measurement arm, (1)

PCMA's offer adaptable measurement systems for multiple applications, such as dimensional analysis, computer-aided design (CAD) inspection, reverse engineering, etc. (2). The main limitation of PCMA's is the effect of manual operation due to PCMA's current joint design, which can introduce positional instability and dynamic structural deformation during measurement due to the joints' inability to hold position (3). Manual operation can also inhibit calibration procedures as it is required that the PCMA must be held still during the procedure (4).

As a result, designing a more effective joint with a position-holding mechanism would improve the design of a PCMA and is a good practice of safety. By designing the joint in this manner, error due to manual operation will be minimised, and kinematic stability and accuracy will be improved.

1.2 Aim and Objectives

The aim of this project is to implement a position-holding mechanism on a PCMA without jeopardising its accuracy and ease of movement.

The objectives of the project are:

1. To generalise the conventional Denavit-Hartenberg(D-H) model derived by Professor Schreve and Mr. Hall (5).
2. To design, build and test key components of the joint to assess the joint design's mechanical feasibility.
3. To integrate the test results into the D-H model and assess feasibility of the joint design.

1.3 Scope

In the process of determining D-H parameters, user manuals and available schematics will be used for information extraction. In addition to this, due to project timelines, physical implementation of the proposed design will be restricted to constructing critical subsystems necessary to validate the core mechanical, electrical, and software functionality.

1.4 Motivation

PCMA's are becoming more popular than traditional coordinate measuring machines (CMM) for in-line¹ and on-site industrial measurement (1). They have broad applicability across multiple industries such as aerospace, automotive, and heavy manufacturing, where having a portable measurement instrument decreases handling time, production delays, and revision of work.

Despite their growing popularity, the PCMA's revolute joints must satisfy competing demands, including:

- Tight run-out tolerances for spatial accuracy.
- Sufficient holding torque to keep the arm stationary when required.
- Internal cable routing.

This has proved difficult since no commercially available joint designs have been found to meet these demands.

¹In the context of PCMA's, "in-line" refers to performing quality inspections and measurements directly on the production shop floor, instead of a temperature-controlled quality laboratory, (1).

Additionally, it has been established that angular position errors at each joint propagate through the kinematic chain, which affects end-effector's accuracy. This proves that joints have a measurable impact on the PCMA's accuracy (6). This project addresses the knowledge gap by deriving joint specifications from a generalised model based off of an error model that was developed by a Stellenbosch Alumnus, Leon Hall (5). This will produce a validated joint and testing methodology to support future PCMA development.

1.5 Use of AI Tools

Gemini is an artificial intelligence (AI) tool that will be used to check grammatical errors and sentence structure as well as LaTeX formatting for the report.

1.6 Report Overview

This section follows the following structure:

- Chapter 1 introduces the project background, aim and objectives, scope, and motivation for project.
- Chapter 2 presents a literature review discussing sources relevant to PCMA joint design, position-holding technology, kinematic modeling, error analysis, and PCMA testing.
- Chapter 3 details the PCMA's requirement analysis of the PCMA joint.
- Chapter 4 discusses concept development for the joint, in which a final concept is chosen and refined.
- Chapter 5 describes the design analysis of the chosen concept, with regards to its mechanical, electronic, and software design.

Chapter 2

Literature review

2.1 PCMA Context

As earlier stated, PCMA's are precise, handheld metrology instruments used to capture 3D measurements directly from physical objects. These devices are built with components that operate similarly to an elbow, forearm, and wrist (6). Typically, an arm operates with six to eight degrees-of-freedom (DOF) to reach inaccessible areas. The measurement accuracy of a PCMA relies heavily on high-precision probe as an end-effector and high-resolution angular encoders located at each rotating joint (6). When an operator touches an object through the probe, the arm's movement is tracked in real time by the angular encoders (1). These measurements are used calculate and plot the exact 3D coordinates of the object within the PCMA's reference system (1). More advanced PCMA models may include internal temperature sensors to compensate for environmental temperature changes. They may also incorporate laser line scanners for non-contact, high-frequency data capture (6).

It is noted that PCMA's offer significant advantages in flexibility and portability, however, their measurement accuracy is dependent on a rigorous calibration procedure (7). This procedure is used to identify D-H parameters (which will be discussed further Section 2.2) and account for them to improve the accuracy of the PCMA, allowing the PCMA's internal software to compensate for error in end-effector measurement results (3). Without proper calibration, the inherent positioning error in the multi-joint arm would exacerbate the arm's measurement uncertainty. This would result in significantly less accuracy in PCMA's (3).

A few authors have highlighted several methodologies and challenges in PCMA calibration to address structural limitations. Alvarez et al. explain that calibration utilises D-H method to model geometric coordinate transformations between consecutive reference systems. They make use of non-linearisation algorithms to minimise measurement errors. Their research features novel data acquisition methods that use laser tracker multilaseration to achieve accurate nominal coordinates across the arm's working volume. In addition to this, Alvarez et al. note that human factors introduce non-uniform contact forces and variable

operating trajectories due to the manual operation. Through finite element analysis (FEA) and multibody dynamics, the demonstrate that human-driven forces induce elastic, dynamic, structural deformations within the arm during probing. Due to linearity of these deformations, Alvarez et al. conclude that calibration models must account for applied pressure in order to improve overall accuracy. Ibaraki and Saito note that overall measurement uncertainty of PCMA is aggravated by angular position errors within the rotary axes, which often stem from graduation errors¹ of the rotary encoder scale. To minimise this effect, Ibaraki and Saito employ the single point articulation test (SPAT) test and modify it with 3D displacement sensors to dynamically identify these angular errors.

Despite these advancements, the proposed solutions of the aforementioned authors, including Alvarez et al. and Ibaraki and Saito, focus solely on calculating and compensating for the error computationally. There is insufficient exploration into mechanical stabilisation techniques to address the positional error observed. Such an approach would prevent structural elastic deformations at the source, instead of relying on post-measurement software correction.

2.2 The Denavit-Hartenberg (D-H) Model

A Denavit-Hartenburg (D-H) model is a transformation matrix that models an end-effector's position and orientation relative to the system of articulate linkages it is connected to (5). This is done by relating the end-effector's reference coordinate system to the last linkage's reference coordinate system through the matrix (5). Hall and Schreve (5) conducted a full investigation of measurement error in PCMA, which also accounted for spatial error using a modified D-H model. The motivation behind this investigation was that PCMA manufacturers did not adopt a common standard that properly governed their products' accuracy (5). It was found that there was a discrepancy in advertised PCMA accuracy to actual PCMA accuracy as manufacturers did not account for spatial dependency in their product (5).

The first step of the project was derive a D-H model, based on a six DOFs PCMA. This PCMA was modelled as a serial kinematic chain due to its architecture. The end effector's position is seen in Equation 2.1:

$$T_6^0 = A_1 \cdot A_2 \cdot A_3 \cdot A_4 \cdot A_5 \cdot A_6 \quad (2.1)$$

wherein, each A_i models the position of each relative to the previous joint's

¹Graduation inaccuracy refers to microscopic physical imperfections in the spacing, placement, or width of the measurement lines (graduations) inscribed on a rotary encoder's internal scale.

reference coordinate system, as seen in Equation 2.2:

$$A_i = \begin{bmatrix} \cos \theta_i & -\sin \theta_i \cos \alpha_i & \sin \theta_i \cos \alpha_i & a_i \cos \theta_i \\ \sin \theta_i & \cos \theta_i \cos \alpha_i & -\cos \theta_i \sin \alpha_i & a_i \sin \theta_i \\ 0 & \sin \alpha_i & \cos \alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (2.2)$$

where a_i is the joint length, θ_i is the rotational angle, α_i is the twist angle, and d_i is the joint offset. These are known as D-H parameters. Together, in Equation 2.2 shown above, they define how a link translates and rotates from one joint's frame of reference to the next.

The D-H model relevant to the joint-design project modelled a Faro Quantum X arm as seen in Figure 2.1 below.

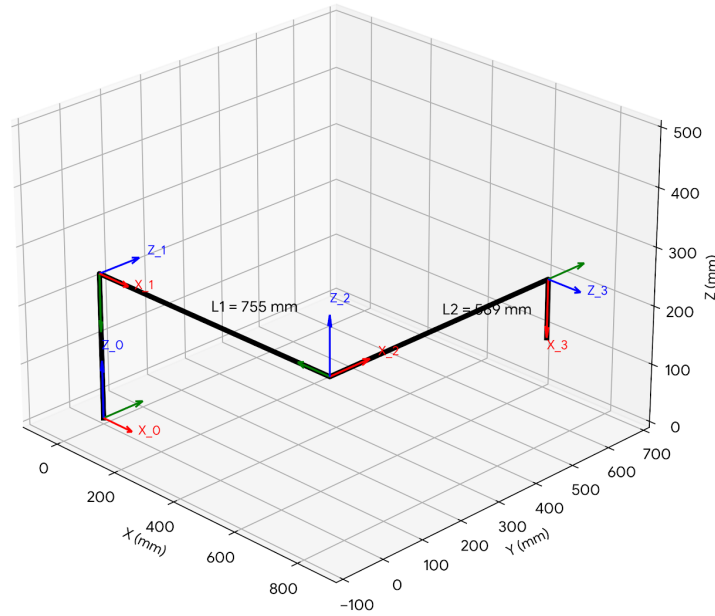


Figure 2.1: D-H Model of the Faro Quantum X Arm

The full derivation of the D-H model is documented in Appendix A.

2.3 The Error Propagation Model

From the derivation of the D-H model, the error propagation model could be derived. Hall and Schreve (5) explain that the model is a mathematical depiction of how known uncertainties move through an articulated system and how those uncertainties effect the output position of the end-effector. The output-position uncertainty is calculated using Equation 2.3 below:

$$\Lambda_F = J(x) \cdot \Lambda_x J(x)^T \quad (2.3)$$

, where $J(x)$ is the jacobian matrix of the end-effector's position and Λ_x is a matrix of the known uncertainty of the encoder at each respective joint. The derivation for Equatio 2.3 begins with a Taylor series expansion, which approximates the articulated system's behaviour near a known point using first-order terms only, which produces the Jacobian matrix. Higher-order terms are negligibly small. The Jacobian matrix is then used to scale the known uncertainties Λ_x and calculate the unknown uncertainty of the end-effector, Λ_F .

The limitation found to this model was that the uncertainty of end-effector could be calculated for only one pose of the articulated system at a time. For the purposes of this project, the error propogation model was generalised by taking a sample of the working volume of the Faro Quantum X arm and computing the uncertainty of the end-effector across the sampled work volume. The software used to conduct these computations was MATLAB, which was also used to plot the distribution of uncertainty across a sampled workspace, which can be seen in Figure 2.2

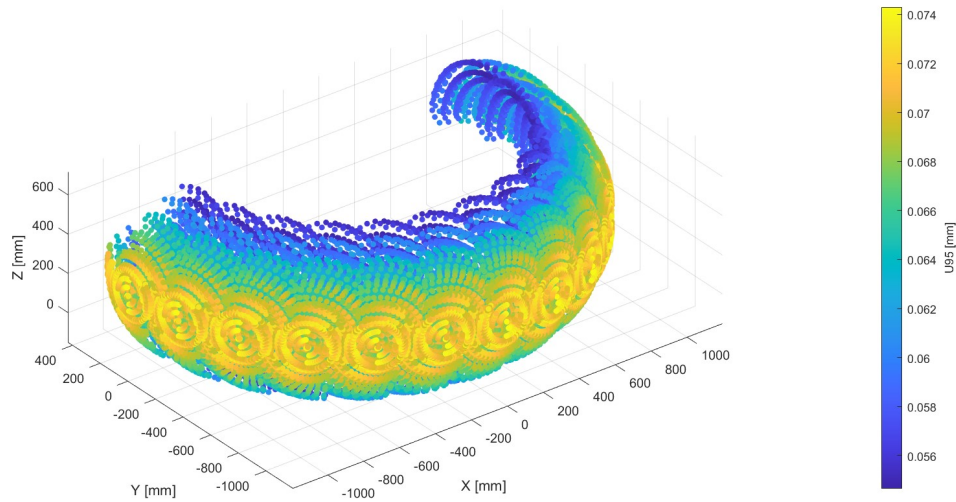


Figure 2.2: Uncertainty Distribution Across Sampled Workspace

The dark blue regions show where the arm is most accurate, which is at approximately 56 microns of uncertainty. This occurs when the arm is folded closer to its base. The bright yellow regions show where the arm is least accurate, up to 74 microns. This occurs at the outer edges of the workspace where the arm is fully extended.

The full capabilities of the code also include a section in which certain joints can be locked and the effect of the locked joints can be computed and plotted on a graph similar to the one found in Figure 2.2. This means that the software can also be used for the future developement of a full PCMA arm with position-holding implementation at optimal joint locations.

2.4 Prior Knowledge

2.5 Testing Design

Chapter 3

Requirement Analysis

The objective of the requirement analysis is to establish clear expectations of the PCMA's joint design for the purposes of concept development.

3.1 Identification of Stakeholders

The first step in the requirement analysis process is to identify stakeholders from the operational and support life cycle phases of the project. Stakeholders will be classified into the following categories, as defined below:

- Primary stakeholders directly interact with the system while it is in operation.
- Secondary stakeholders contribute to the development and functioning of the system.
- Indirect stakeholders rely on or are affected by the system, but do not interact with the system directly.

The stakeholders relevant to the project are defined in Table 3.1 below.

Table 3.1: Identified stakeholders and their classifications

Stakeholder ID	Description	Classification
SID_01	Supervisor of the project	Primary
SID_02	Students researching PCMA development	Secondary
SID_03	Companies invested in PCMA development	Indirect
SID_04	Mechanical workshop staff	Indirect
SID_05	Mechatronics laboratory managers	Indirect

3.2 System Context

The next step is to define the narrower system of interest (NSoI) and the wider system of interest (WSol) of the project¹, (8) for the purposes of this project, the joint was identified as the NSoI and WSoI is the NSoI with a reliable power supply and an artefact with which the joint will be tested. Figure 3.1 illustrates the system context, with the NSoI shown as a "black box", (8).

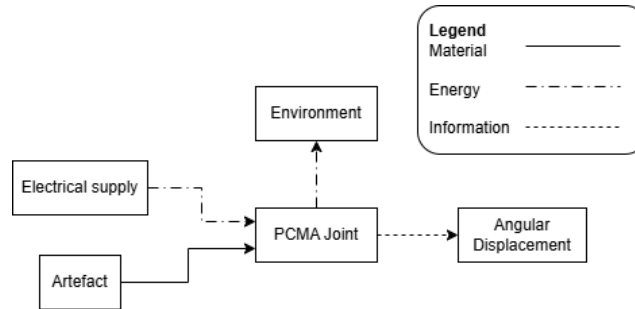


Figure 3.1: System Context

3.3 Design Independent Parameters

Next, the design independent parameters (DIPs) are defined. DIPs influence the system during the operational and support life cycle stages, but are not controllable nor are part of the system being defined, (8). The DIPs of the joint design are detailed in Table 3.2 below.

Table 3.2: Design Independent Parameters

DIP_ID	Description	Expected range
DIP_01.1	Electrical supply	12 V - 24 V
DIP_01.2	Electrical supply	12 A maximum
DIP_01.3	Electrical supply	Direct current
DIP_02.1	Artefact	Compliant with ISO 10360-12:2016
DIP_02.2	Artefact	20 points reference points
DIP_03.1	Environment	Laboratory ambient temperature: 0 °C - 35 °C
DIP_03.2	Environment	Laboratory ambient humidity: 20 % - 80 %
DIP_03.3	Project budget	R 6000

¹The NSoI is the system being designed and the WSoI includes the NSoI and elements that the NSoI interacts with, (8).

3.4 Functional Requirements

In this section, the primary and top-level derived functional requirements (FRs) for the joint design are defined. The only FRs that are considered are only associated with joint's operation. FRs are categorised into primary and derived FRs, in which:

- Stakeholders are directly concerned with primary FRs.
- Derived FRs are required so that the system can perform primary FRs.

As seen in Table 3.3 below, the FRs for the joint design are listed. In Figure 3.2,

Table 3.3: Table 3: Functional Requirements

FR ID	Description	Primary / Derived
FR_01	Articulate output link	P
FR_02	Hold position	P
FR_03	Measure angular displacement	P
FR_04	Store angular displacement data	D
FR_05	Release position	P

the functional requirements of the joint are depicted below.

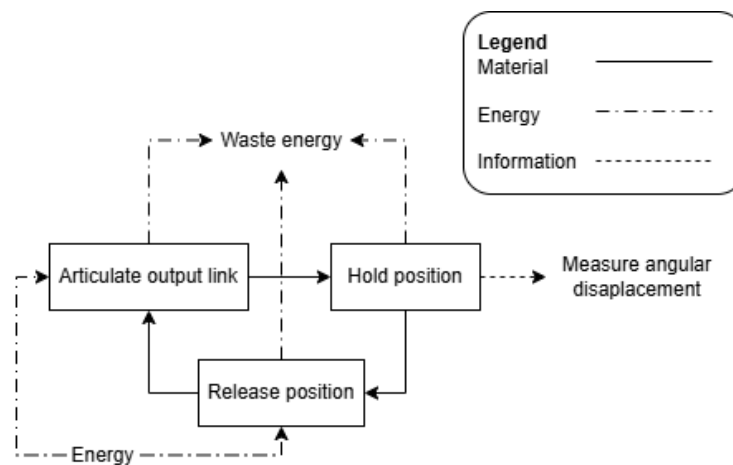


Figure 3.2: Top Level Functional Decomposition

3.5 Technical Performance Measures

This section defines the technical performance measures (TPMs) which will be used to determine how well the joint design (NSoI) performs the required

functions². The TPMs, listed in Table 3.4, were derived from the results obtained from the generalised error-propagation model.

Table 3.4: Technical Performance Measures

TPM ID	Related FR	Description	Target range
TPM_01	FR_01, FR_05	Operational articulation angle	90°
TPM_02	FR_01, FR_05	Run-out tolerance	1.571×10^{-3} rad
TPM_03	FR_02	Holding torque	7.725 Nm
TPM_04	FR_03, FR_04	Encoder accuracy	1000 PPR
TPM_05	FR_01, FR_02, FR_05	Joint life cycle	10^6 life cycles

²TPMs are measurable and unambiguous parameters that can be influenced by designing the joint

Chapter 4

Concept Development

4.1 Concept Generation

The generalised error propagation model was used to derive the TPMs in the development of the joint design. These TPMs guided the research into various mechanical solutions, and different mechanisms were evaluated for implementation to ensure the final design could achieve the required TPMs. Three concepts were generated and evaluated against the FRs and the TPMs, as presented in this section.

4.1.1 Concept One: PCMA Joint with a Locking Dial

Figure 4.1 below shows a PCMA joint with the following internal modifications to implement position-holding capabilities. In the center of the shaft, is a spacer between angular contact bearings, which will allow the shaft to rotate with minimal run-out. Bellville washer and preload nuts are used to secure the bearings in place. On one end, the rotary encoder's shaft is connected to the joint's shaft via a rigid coupling and the encoder's body is held secure by the joint housing. This is done to ensure that the relative angular displacement of the shaft with respect to the joint housing is measured. The joint housing is held stationary by the base link and the joint shaft is rotated by the output link. The knob on the other end of the shaft is meant to loosen and/or tighten the preload nuts against the bearings to lock or release the shaft for the output link's rotation.

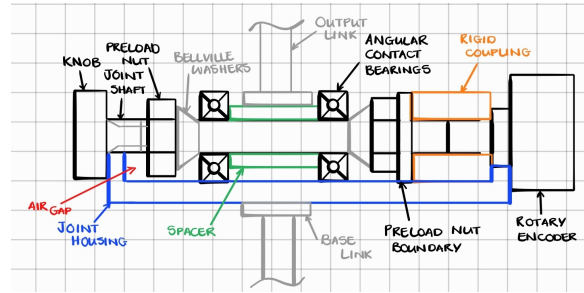


Figure 4.1: PCMA Joint with Locking Dial

4.1.2 Concept Two: PCMA Joint with a Counterbalance Spring

Figure 4.2 illustrates a PCMA joint with external modifications to implement position-holding. The counterbalance spring produces a lifting force to the upper link via the upper and lower rods. Each rod is pin-connected to a cuff that can be fixed in position along the length of either link using a set screw, thereby holding the position of the joint. The

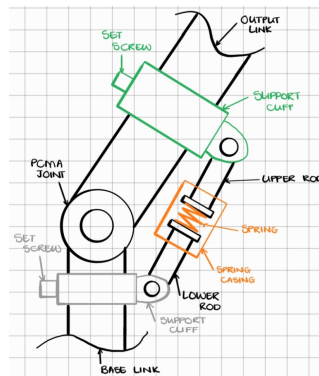


Figure 4.2: PCMA Joint with Counterbalance Spring

4.1.3 Concept Three: PCMA Joint with an Electromagnetic Brake

Figure 4.3 depicts a PCMA with internal modifications to implement an electromagnetic brake. In the center of the shaft, two angular contact bearings are fitted in place with the bearing housing and pressed against a shoulder of the shaft by preload nuts and washers. On one side of the shaft, the rotary encoder's shaft is connected to the joint's shaft via a rigid coupling and the encoder's body is held stationary by the joint housing. The base link supports the joint housing. The electromagnetic brake sits on the other side of the shaft and is also held stationary by the joint housing, however, the brake disc is free to rotate on the joint shaft. When the brake is energized, the air gap between the brake disc and friction plate opens up and the shaft is free to be rotated by the output link.

When the brake's power supply is cut off, the air gap closes and the joint is held still.

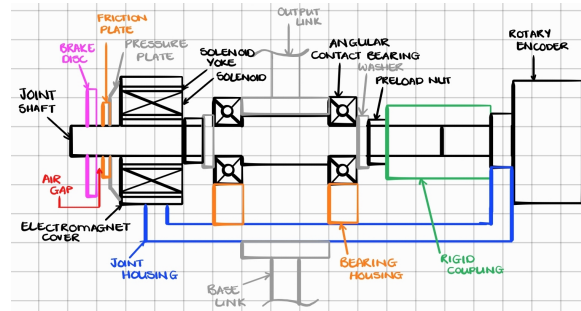


Figure 4.3: PCMA Joint with Electromagnetic Brake

4.2 Concept Evaluation

The concepts presented in the previous section were determined to be feasible. A qualitative evaluation of the concepts is to be conducted as the generated concepts are still solution neutral and cannot undergo a full quantitative comparison. This section compares the concepts in terms of the following evaluation criteria (EC):

- EC1 Load capacity: The ability of the joint design to carry the required holding torque (TPM_03).
- EC2 Operational articulation angle: The ability of the joint design to achieve the full required sweep without interference (TPM_01).
- EC3 Positional stability: The ability of the joint design to maintain position when the position-holding technology is in use (TPM_02).
- EC4 Component availability: The degree to which all required components are available in the mechanical workshop.
- EC5 Construction simplicity: The number of distinct components and the ease of assembly and alignment during fabrication.

EC1 was derived from TPM_03 and determined to be the most important EC as the joint will be ineffective if it is unable to achieve the holding torque dictated by TPM_03. EC2 was derived from TPM_01 as the chosen joint design must not inhibit the PCMA's existing range of motion once implemented. EC3 was derived from TPM_02 as any internal sources of error must be minimised to maximise the joint design's ability to maintain and/or improve the PCMA's accuracy. EC4 and EC5 was included as a practical design consideration given the manufacturing capabilities of the mechanical workshop and the project's timeline and budget.

Table 4.1 below, presents a comparison of the generated concepts. To rate the concepts, one of them is marked as the DATUM and the other two are awarded a +1 if the other concept is stronger or –1 if it is weaker. Each rating is motivated in the table as well.

Table 4.1: Concept Evaluation

Criterion		Concept 1: PCMA Joint with a Locking Dial			Concept 2: PCMA Joint with a Counterbalance Spring (DATUM)		Concept 3: PCMA Joint with an Electromagnetic Brake	
#	Description	Wt.	Rtg	Motivation	Dat	Motivation	Rtg	Motivation
EC1	Load capacity	5	-1	The capability of the design achieving the exact holding torque required is entirely dependent on the operator's strength and consistency, otherwise the joint design is highly susceptible to slipping under load.		Datum: The set screw mechanism on the support cuffs provides a friction-based hold, which establishes the baseline for evaluating the holding torque FR.	+1	An adequately specified electromagnetic brake can consistently deliver a static holding torque that meets the holding torque FR. This removes human reliability (in comparison with concept 1) and ensures reliable load capability when the joint is powered.
EC2	Operational articulation angle	5	+1	The locking mechanism is entirely internal. Because there are no external linkages, the joint can achieve its full required 90° sweep without any physical interference.		Datum: The external arrangement of the upper and lower rods, including the spring casing, limit the joint's movement. This obstruction serves as the baseline limitation for the required articulation sweep.	+1	Similarly to concept 1, the electromagnetic brake is neatly integrated with the joint shaft, which allows for uninhibited articulation
EC3	Positional stability	5	-1	Manually twisting the dial introduces asymmetric axial and radial loads onto the joint shaft. This can cause deflections that violate the run-out tolerance FR and degrade the PCMA's spatial accuracy.		Datum: Fixing the joint's position by tightening set screws on the external cuffs introduces off-center loads to the joint shaft. This represents the baseline for positional deflection and run-out error.	+1	The electromagnetic brake applies a uniform, symmetric clamping force between the brake disc and friction plate. This eliminates manual influence during the locking process, as well as, minimises parasitic deflection and protects the system's positional accuracy.
EC4	Component availability	5	+1	The required components (threaded shafts, preload nuts, bellville washers) are items that either readily available or easily machined in the university workshop.		Datum: The design relies on standard mechanical items such as springs, rods, and set screws, which defines the baseline for material availability due these components being accessible in the mechanical workshop.	0	While electromagnetic brakes are commercially available, specific ordering is required, lead times, and budget allocation compared to the raw materials used in the datum design.

Table 4.1 – Continued from previous page

Criterion			Concept 1: PCMA Joint with a Locking Dial		Concept 2: PCMA Joint with a Counterbalance Spring (DATUM)		Concept 3: PCMA Joint with an Electromagnetic Brake	
#	Description	Wt.	Rtg	Motivation	Dat	Motivation	Rtg	Motivation
EC5	Construction simplicity	5	0	The mechanical assembly is relatively straightforward and on the level of simplicity as the datum design. However, achieving accurate bearing alignment under varying manual requires extra care.		Datum: Assembling the external support cuffs, rods, and pin-connections require a standard level of manual alignment and fabrication, setting the baseline for manufacturing complexity.	-1	This is the most complex design to implement. It requires precise mechanical calibration of the encoder, as well as integration of electrical wiring and control circuitry for the solenoid.
Number of + ratings			2				3	
Number of - ratings			2				1	
Overall total			0				2	
Weighted total			0		0		+10	

4.3 Concept Selection

Based on the qualitative evaluation presented under Section 4.2., Concept 3 is selected as the final design, as shown in Figure 4.1, under Section 4.1.3.

The motivation behind selecting this concept is its superior performance across the most critical functional criteria, specifically load capacity (EC1) and positional stability (EC3).

The electromagnetic brake provides a consistent and reliable static hold, in contrast to the manual locking dial which relies heavily on human reliability and strength. When de-energised, the electromagnetic brake is able to successfully meet the strict holding force requirement by applying a uniform clamping force between the brake disc and friction plate. The uniform clamping eliminates off-center loads and minimises parasitic deflections, thereby preserving the tight run-out tolerances required for the PCMA's spatial accuracy. In addition to this, the electromagnet is integrated internally on the joint shaft, ensuring that the joint design does not impede the joint's physical movement. This allows the arm to achieve the full required 90° operational articulation sweep without interference (EC2).

To confirm that the chosen selected concept is well suited to perform the functional requirements in Table 4.1, Table 4.2 below, identifies the components to fulfilling the respective functions.

Table 4.2: Concept Functional Mapping

Function Number	Function Description	Concept
FR_01	Articulate output link	Energising the electromagnetic brake opens the air gap, allowing the joint shaft to be freely rotated by the output link.
FR_02	Hold position	When the electromagnet is de-energised, the air gap closes and the mechanism applies a static holding force to keep the joint stationary.
FR_03	Measure angular displacement	A rotary encoder is connected to the joint shaft via a rigid coupling, measuring relative rotation while the encoder body is held stationary by the housing.
FR_04	Store angular displacement data	The angular displacement data captured by the rotary encoder is transmitted to the microcontroller, which is part of the system's switching circuitry.
FR_05	Release position	Energising the brake's solenoid retracts the pressure plate from brake disk, releasing the joint for movement.

4.4 Design Refinement

The conceptual design of the electromagnetic joint was refined into a 3D CAD assembly to verify spatial feasibility, component scaling, and integration into the wider system of interest, as seen in Figure 4.4.

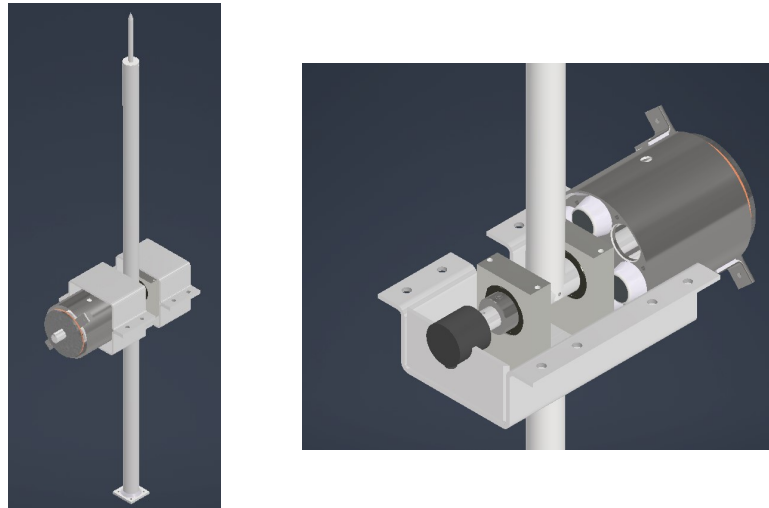


Figure 4.4: Selected Concept Design

The design was modelled to ensure that the main components are correctly sized and positioned to integrate seamlessly within the PCMA's physical constraints. The model also ensures that the means of operation and load paths are clearly defined. This ensures the design does not impede the required 90° operational articulation sweep.

The primary mechanical requirement for the refined mechanism is to deliver a reliable static holding torque of 7.725 Nm. To achieve this, the electromagnetic brake is held stationary by the joint housing, with a loose fit on the same shaft as the rest of the joint. Under normal measurement operation, the electromagnet is energised with an operating current of 12 A. The brake disc is fixed to rotate with the shaft via a key and the friction plate is connected to the pressure tabs on the electromagnetic brake and is held stationary as a result. The force exerted as a result, retracts the pressure tabs and friction plate, thereby creating an air gap between the brake disc and friction plate to allow the shaft to rotate freely in its bearings. The shaft is rotated by a lightweight output link¹, which is connected to the shaft through a dowel pin.

The position-holding capabilities of the brake are activated by shutting off power to the electromagnet. The pressure tabs return to their resting position and force the friction plate against the brake disc. This is how the uniform clamping force is physically implemented to stop the shaft from rotating, thereby holding the joint's position. This configuration ensures uniform application of the holding torque to minimise parasitic radial deflections. This further ensures that the run-out tolerance is maintained for the PCMA's spatial accuracy.

The design also accommodates the necessary sensing and structural support subsystems. A high-resolution rotary encoder is connected to the opposite end of the joint shaft via a rigid coupling. The encoder's body is secured to the joint housing, which encases the entire joint assembly and held stationary by the

¹The output link represents the distal link of the PCMA arm.

base link². This ensures that any angular displacement of the shaft, relative to the stationary joint housing, is accurately captured and transmitted to the control circuitry. The shaft is supported by two roller bearings, which were chosen over angular contact bearings due to their cost-effectiveness and availability in the market in comparison. These bearings are mounted onto the shaft with an interference fit and shaft collars to prevent axial slipping. In addition to this, the shaft was designed with shoulders to maintain spacing, while the bearing housing was integrated into the joint housing to further eliminate axial play.

Below is a parts list that corresponds to the refined joint subsystem in Figure 4.4, excluding standard/trivial fasteners:

Table 4.3: Refined Design Parts List

Item No.	Component Description	Quantity
1	Joint shaft	1
2	Output link	1
3	Roller bearings	1
4	Shaft collars	2
5	Bearing housing	2
6	Electromagnetic brake assembly	2
7	Rigid shaft coupling	1
8	Rotary encoder	1
9	Base link	1
10	Joint housing	1
11	Friction plate	1
12	Brake disc	1

²The base link represents the proximal link of the PCMA arm.

Chapter 5

Design Analysis

The design analysis of the electromagnetic brake joint is divided into four hierarchical layers, including the WSoI which encompasses the testing environment and the PCMA arm, the NSoI representing the joint, its functional subsystems, and the individual components. The physical development of the joint was iterative, however, the resulting architecture is documented in a logical, top-down fashion instead of a chronological sequence. To describe the system's integration, the analysis is split into three domains which constitute the mechanical design, the electronic design, and the software design. The mechanical design outlines the spatial arrangement and loading paths. The electronic design describes the peripheral circuitry and power interfacing. The software design defines the control logic and the state machine controlling the electromagnetic brake.

5.1 Mechanical Design

Figure A.1 below, shows the 3D CAD assembly of the electromagnetic brake produced as part of the design analysis after integration and refinement of the individual subsystem concepts.

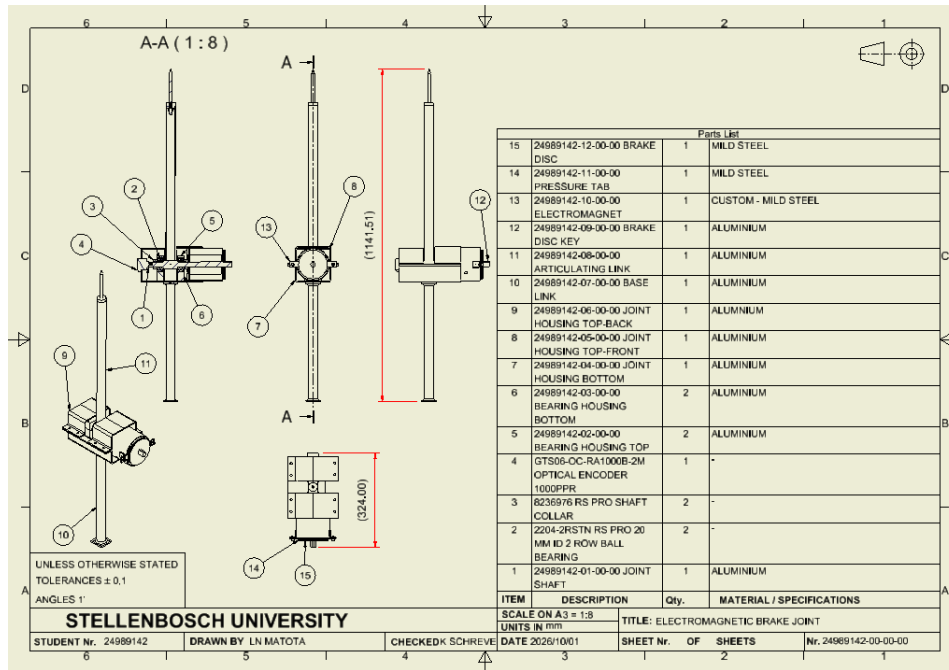


Figure 5.1: Electromagnetic Brake Assembly

The top-level system elements are listed in the drawing's materials list and are marked with balloon numbers in Figure A.1. The different views and dimensions of the engineering drawing depict the position of physical interfaces with elements external to the NSoI and the spatial constraints for the respective system elements inside the NSoI.

The overall configuration uses an electromagnetic brake mounted coaxially on the joint shaft. In the configuration, the base link and the output link are integrated with the internal actuation and sensing components. The joint is supported by two RS PRO 20 mm ID 2-Row ball bearings mounted with an interference fit and secured by RS PRO 32 mm OD one-piece steel shaft collars to prevent axial slipping. The bearing housing is integrated directly into the stationary joint housing. The housing encases the internal mechanisms and is held rigid by the base link. On one of the shaft, the GTS06-OC-RA1000B-2M optical encoder is connected via a Ruland MCLX-20-20-F rigid black oxide steel coupling. A spacer sits on the side of the encoder shaft to compensate for the discrepancy in diameters of the joint, the encoder, and the rigid coupling¹. On the opposite end, the electromagnetic brake assembly interfaces directly with the shaft. The electromagnetic brake consists of solenoids, friction plate (laser-cut by Fabrinox), and a brake disc.

This configuration was chosen because it provides a reliable, uniform clamping force that minimises parasitic deflections and preserves the run-out tolerances necessary for the PCMA's spatial accuracy. Additionally, by housing the brake internally, the design is able to achieve the full 90° operational articulation

¹See Appendix ## for calculations regarding the joint shaft design.

sweep.

To maintain structural integrity and positional stability during locking, the electromagnetic brake was sized to deliver a static holding of 7.725 Nm, corresponding to a electromagntive force of 3550 N. The freebody diagrams (FBDs) and design calculations supporting the mechanical operating conditions of the joint under Appendix ##.

5.2 Electronic Design

The purpose of the switching circuit is to bridge the significant power gap between the high-power electromagnet and the low-power control logic, as well the data management system. The stm32 Nucleo microcontroller outputs a low-current control signal, which is insufficient to directly drive the electromagnetic brake that requires a 12 A operating current at 24 V_DC. This means that peripheral circuitry is required to safely step down power for the logic and allow the microcontroller to switch the electromagnet without sustaining damage. Figure 5.2 below shows a wiring diagram detailing these connections.

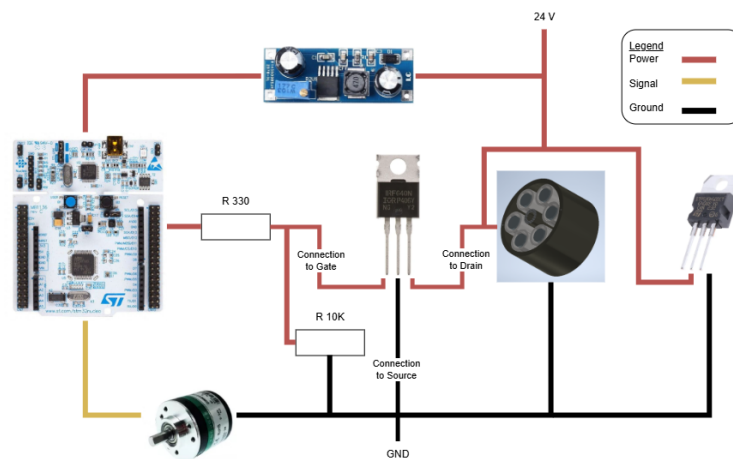


Figure 5.2: Wiring Diagram

To power the logic components, an XL7015 step-down buck converter is implemented. Its purpose is to reduce the 24 V supply to a safe operating voltage suitable for the STM32 microcontroller and the rotary encoder.

To control the energisation of the brake, an Infineon IRF640NPBF N-channel metal-oxide-semiconductor field-effect transistor (MOSFET) is utilised as a low-side switch. The MOSFET acts as a digital relay. It's purpose is to take the control signal from the microcontroller² and use it to safely turn the high-current electromagnet on and off.

²From the microcontroller, one of the pins is configured as a general pin input/output, which outputs control signal typically between 20 mA and 25 mA at an operating voltage level of 3.3 V.

There are supporting components in the switching circuit which are critical and protective of the MOSFET, including the $330\ \Omega$ gate resistor, the $10\ \text{k}\Omega$ pull-down resistor, and the TO-220 flyback diode. The gate resistor is positioned between the microcontroller's signal pin and MOSFET gate. It's used to limit the sudden current flow needed to charge the MOSFET's internal capacitance in the event of switching the electromagnet on or off. This will prevent the microcontroller pin from burning out. The pull-down resistor is connected between MOSFET gate and drain. This allows the pull-down resistor to force the gate voltage to zero whenever the microcontroller is powering up, resetting, or outputting a floating signal. As a result, the MOSFET defaults to an "off" state, keeping the joint safely locked and preventing unintentional energisation. The flyback diode is placed in parallel across the electromagnetic coil. Its purpose is to minimise voltage spikes that could damage the MOSFET. When the electromagnet's current is suddenly cut off, the collapsing magnetic field generates a back electromotive force (EMF) due to its inductance. The back-emf would result in a high-voltage spike that exceeds the MOSFET's breakdown limit causing catastrophic failure. The diode provides a safe, low resistance path to dissipate the voltage spike.

For the sensing subsystem, the GTS06-OC-RA1000B optical rotary encoder is connected into the logic power rails, with its signal lines that were connected directly to the microcontroller to transmit real-time angular displacement data. The associated electrical calculation sizing these components and evaluating the MOSFET's thermal dissipation are detailed in Appendix ##.

5.3 Software Design

In mechatronics systems, such as the electromagnetic brake system, it is necessary to define the interactions between actuators (electromagnetic brake), sensors (encoder), and the control software for reliable operation. The software architecture for the brake system uses a supervisory control and data acquisition (SCADA) hierarchy to manage real-time monitoring and actuation processes. This hierarchy consists of the following levels, as shown in Figure 5.3:

- Level 0 (field devices) consists of the hardware interacting directly with the optical rotary encoder, the IRF640N MOSFET, and the electromagnet brake.
- Level 1 (control logic) is executed by the STM32 Nucleo microcontroller, which acts as input/output processing module.
- Level 2 (supervisory) is handled by the connected computer that receives angular displacement data from the microcontroller for logging and visualisation.

To fulfill the system's FRs, the microcontroller's pinout is configured to interface with the Level 0 devices. The rotary encoder's Phase A and Phase B signal lines are connected to pins PA0 and PA1. These pins are routed internally to the

STM32's hardware timer (TIM2). The encoder outputs a quadrature pulse³, which allows the system to determine rotational position and direction of the shaft based on which phase leads the other. By utilising X4 encoding, which is when pulses are counting on the rising and falling edge, for both channels, the 1000 pulse per revolution (PPR) encoder yields 4000 distinct counts per 360-degree rotation.

By operating TIM2 in Encoder Mode, the microcontroller is able to use its dedicated hardware circuitry to autonomously track the high-frequency quadrature pulses and update a 32-bit memory register entirely in the background. This type of tracking eliminates the need for CPU-intensive interrupts, which would force the processor to halt execution of the code to count thousands of pulses per second. As a result, the CPU is freed up to properly manage other tasks. Additionally, the MOSFET gate is driven by pin PC2, which is configured as a digital output. Pin PC13 is configured as a hardware interrupt (EXTI), which is tied to the user input button. Lastly, pins PA2 and PA3 are configured for USART2 serial transmission to communicate the level 2 supervisory computer.

The control strategy implemented for the electromagnetic brake is an open-loop, bang-bang controller. Due to the binary mechanical requirement⁴, duty cycle control is unnecessary. As shown in Figure 5.3, the system initialises into a locked state by defaulting PC2 low, thereby creating a fail-safe protocol.

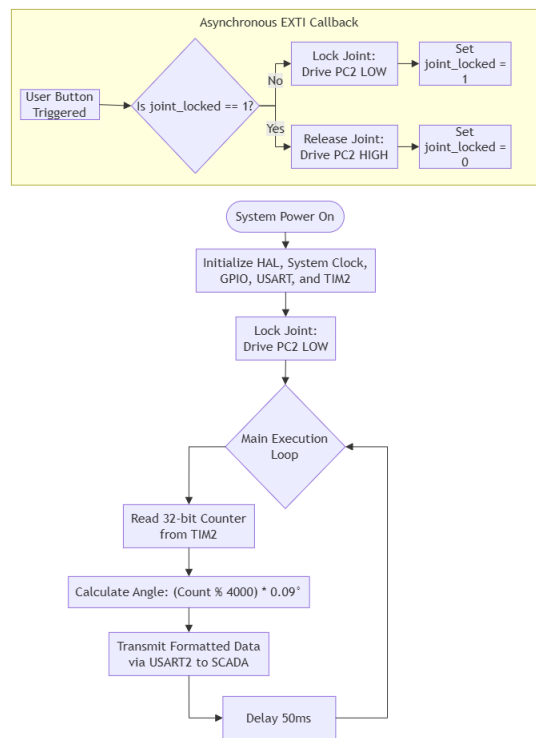


Figure 5.3: Software Design of the PCMA Operation

³A quadrature pulse is a signal that consists of two identical square waves offset by 90°.

⁴The joint must be locked in placed, or released, free to rotate.

The main loop reads the 32-bit counter from TIM2, converts the raw pulse count into an angular displacement into degrees, and transmits this data at 20 Hz to the supervisor computer via USART2.

To manage the energisation and de-energisation of the electromagnet, the software uses the external interrupt on PC13. When the user presses the button, the interrupt service routine (ISR) pauses the main loop and toggles the joint's variable (`joint_locked`) state. If the variable goes high (`joint_locked == 1`), the controller drives the MOSFET gate high (3.3 V). This energises the solenoid and opens the air gap. Due to the asynchronous execution of the interrupt, the user does not experience lag when manually operating the arm. This is done to ensure a seamless human machine interface (HMI) between the mechanical operation and the software control state.

Chapter 6

Design Evaluation and Testing

6.1 Prototype Construction

As stated in the project scope, under section 1.3, the construction of the joint design was restricted to critical subsystems required to validate its core operating functions. As a result, instead of building the full-scale joint assembly, a scaled-down prototype of the primary electromagnetic brake was constructed.

Due to project constraints, including an extended theoretical design phase, material availability and manufacturing facilities of the mechanical workshop, the physical build was adapted to align with these capabilities.

6.1.1 Mechanical Design of Prototype

The resulting design implementation of the prototype resulted in a four-solenoid configuration, wound with copper wire with a 0.28 mm diameter. This specific wire has a significantly reduced maximum current limit compared to the copper wire required for the 6 A rating specified in the theoretical design.

6.1.2 Electrical Design of Prototype

6.1.3 Software Design of Prototype

6.2 Safety and Performance Testing

6.2.1 Holding Force Verification

6.2.2 Electrical Safety and Thermal Limits

Chapter 7

Conclusions

Appendix A

Design Calculations

A.1 Denavit-Hartenberg Model

The D-H model derived for the project is based off of the FaroArm QuantumX arm. The arm has six-DOF and will be modelled as a serial kinematic chain. The first step is to determine the D-H parameters of the PCMA. Note that the nominal D-H parameters of the FaroArm are not published as they are stored in the device's internal electrically erasable programmable read-only memory (EEPROM) firmware or stored within its proprietary device drivers. However, they can be derived from physical dimensions in the product specifications document due to the analogous structure of the Faro QuantumX family (9). The specific length chosen was the three-meter arm and its D-H parameters are tabulated under Table A.1 below:

Table A.1: Denavit-Hartenberg parameters for the FaroArm model

Link i	a_i [mm]	α_i	d_i [mm]	Notes
1	0	+90°	283	Offset as a result of the base.
2	0	-90°	755	Length of link 1 from position sensitive detector (PSD)
3	0	+90°	0	Intersecting elbow axes ¹
4	0	-90°	569	Length of link 2 from PSD
5	0	+90°	0	Intersecting wrist axes ²
6	0	-90°	220	End-effector length

The D-H parameters are:

- a_i is the perpendicular distance between two consecutive joints' axes of rotation.

¹Physically, this represents the twist and hinge encoders of the elbow working together as a multi-axis pivot, instead of being separated by a distinct tubular segment.

²There is a physical housing that connects the encoders, which reduces the link length to zero, creating a joint-like behaviour before the end-effector offset.

- α_i is the angle between two consecutive joints' axes of rotation and can be assumed through deductive reasoning.
 - $+90^\circ$ means the joint axes are perpendicular to each other in a positive orientation.
 - 0° means the joint axes are parallel to each other.
 - -90° means the joint axes are perpendicular to each other in a negative orientation.
- d_i is the perpendicular distance between two consecutive joints' axes of rotation.

Note that there is a rotational angle, θ_i , which is the relative rotation between joints with respect to the previous joint. This θ_i is a real-time variable measured by the encoder at each joint. The next step is to derive the transformation matrix which determines the end-effector's coordinates relative to joint i , as seen in Equation A.1:

$$\begin{aligned}
 A_1 &= \begin{bmatrix} \cos \theta_1 & -\sin \theta_1 \cos \alpha_1 & \sin \theta_1 \sin \alpha_1 & 0 \\ \sin \theta_1 & \cos \theta_1 \cos \alpha_1 & -\cos \theta_1 \sin \alpha_1 & 0 \\ 0 & \sin \alpha_1 & \cos \alpha_1 & 283 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 A_2 &= \begin{bmatrix} \cos \theta_2 & -\sin \theta_2 \cos \alpha_2 & \sin \theta_2 \sin \alpha_2 & 0 \\ \sin \theta_2 & \cos \theta_2 \cos \alpha_2 & -\cos \theta_2 \sin \alpha_2 & 0 \\ 0 & \sin \alpha_2 & \cos \alpha_2 & 755 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 A_3 &= \begin{bmatrix} \cos \theta_3 & -\sin \theta_3 \cos \alpha_3 & \sin \theta_3 \sin \alpha_3 & 0 \\ \sin \theta_3 & \cos \theta_3 \cos \alpha_3 & -\cos \theta_3 \sin \alpha_3 & 0 \\ 0 & \sin \alpha_3 & \cos \alpha_3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 A_4 &= \begin{bmatrix} \cos \theta_4 & -\sin \theta_4 \cos \alpha_4 & \sin \theta_4 \sin \alpha_4 & 0 \\ \sin \theta_4 & \cos \theta_4 \cos \alpha_4 & -\cos \theta_4 \sin \alpha_4 & 0 \\ 0 & \sin \alpha_4 & \cos \alpha_4 & 569 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (A.1)
 \end{aligned}$$

$$A_5 = \begin{bmatrix} \cos \theta_5 & -\sin \theta_5 \cos \alpha_5 & \sin \theta_5 \sin \alpha_5 & 0 \\ \sin \theta_5 & \cos \theta_5 \cos \alpha_5 & -\cos \theta_5 \sin \alpha_5 & 0 \\ 0 & \sin \alpha_5 & \cos \alpha_5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_6 = \begin{bmatrix} \cos \theta_6 & -\sin \theta_6 \cos \alpha_6 & \sin \theta_6 \sin \alpha_6 & 0 \\ \sin \theta_6 & \cos \theta_6 \cos \alpha_6 & -\cos \theta_6 \sin \alpha_6 & 0 \\ 0 & \sin \alpha_6 & \cos \alpha_6 & 220 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The D-H parameters are input into the error propagation model to determine the transformation matrices in Equation 2.2 above. These calculations can be cumbersome if done by hand, however, Matlab was chosen to complete them which will be discussed in the following Appendix A.2.

A.2 Error-Propagation Model

The purpose of the error-propagation model is to determine the position uncertainty distribution of the PCMA's end-effector across a sampled workspace. As shown in Figure A.1 below, the algorithm receives the D-H parameters of the chosen PCMA which is the FaroArm QuantumX arm, specifically, the 3 m arm.

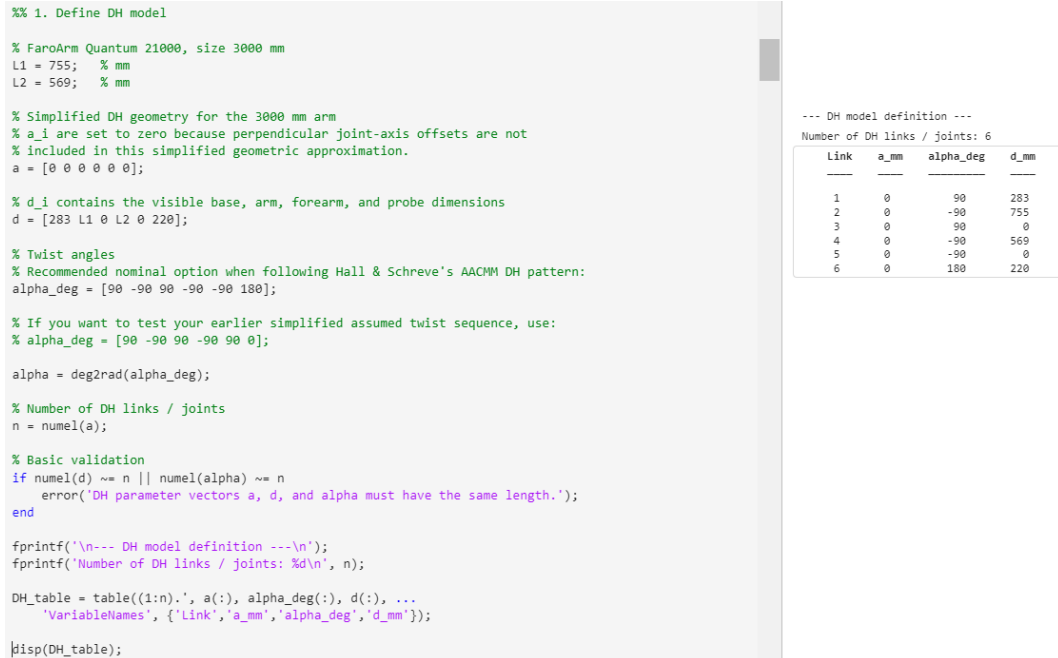


Figure A.1: Error-Propagation Variable Input

In the next step, the encoder variance is input into the algorithm as the identity matrix, Λ_θ . This is done under the assumption that the encoders are independent and equal variance. The covariance used is 6 arc-sec based off of industry standard, as shown in Figure A.2.

```
%% 2. Define encoder covariance

% Nominal encoder standard deviation
% This value is a scaling assumption. It should later be calibrated using
% SPAT, repeatability, or experimental FaroArm data.
sigma_theta_arcsec = 6;
sigma_theta = arcsec2rad_local(sigma_theta_arcsec);

% Independent equal-variance encoder covariance matrix
Lambda_theta = (sigma_theta^2) * eye(n);
```

--- Encoder covariance ---
Assumed encoder standard deviation = 6.000 arc-sec
Assumed encoder standard deviation = 2.908882e-05 rad

Figure A.2: Error-Propagation Encoder Covariance Input

Next, the joint sweep for each joint determined, such that an angle range is chosen for each joint and divided into a specified number of samples, as shown in Figure A.3 below.

```
%% 3. Define joint sweep / locked-joint configuration

% Each cell represents one joint.
% Each cell define the set of arm configurations (poses) at which the
% uncertainty model is evaluated.
% Each entry corresponds to one joint:
% ~ A vector -> joint is swept
% ~ A scalar -> joint is locked (0)

theta_lists_deg = {
    ... theta1: base rotation
    ... Base rotation sweeps from left to right across the workspace
    linspace(-90, 90, 13)...
    ... theta2: shoulder
    ... Shoulder operates within a mid-range motion band
    linspace(10, 80, 10)...
    ... theta3: elbow
    ... Elbow allows extension and contraction of the arm
    linspace(-60, 60, 10)...
    ... theta4: wrist
    ... Wrist joints vary within a smaller angular range
    linspace(-45, 45, 7)...
    ... theta5: wrist
    linspace(-45, 45, 7)...
    ... theta6: wrist / probe roll
    linspace(-45, 45, 7)
};
```

--- Joint sampling setup ---
Total number of poses to evaluate: 445900

theta1 swept from -90.000 deg to 90.000 deg using 13 values
theta2 swept from 10.000 deg to 80.000 deg using 10 values
theta3 swept from -60.000 deg to 60.000 deg using 10 values
theta4 swept from -45.000 deg to 45.000 deg using 7 values
theta5 swept from -45.000 deg to 45.000 deg using 7 values
theta6 swept from -45.000 deg to 45.000 deg using 7 values

Figure A.3: Joint Sweep Sample

These angle samples translate into a sample of the working volume. It is within this sample that every pose determined by the number of angle samples from the Figure A.3 above, that the position uncertainty distribution can be determined, using the error propagation model, as seen in Figure A.4 below.

```

%% 5. Evaluate uncertainty over all generated poses

fprintf('\n--- Evaluating poses ---\n');

% Length of each joint list
list_lengths = zeros(1,n);
for i = 1:n
    list_lengths(i) = numel(theta_lists_deg{i});
end

% Variables for best/worst tracking
maxU95 = -inf;
minU95 = inf;

worst_pose_deg = zeros(1,n);
best_pose_deg = zeros(1,n);

worst_position = zeros(3,1);
best_position = zeros(3,1);

```

(a)

```

for idx = 1:N

    % Convert linear index into subscript indices for each joint list
    subs = cell(1,n);
    [subs{:}] = ind2sub(list_lengths, idx);

    % Build joint vector for this pose
    theta_deg_i = zeros(1,n);

    for j = 1:n
        theta_deg_i(j) = theta_lists_deg{j}(subs{j});
    end

    theta_i = deg2rad(theta_deg_i);

    % Evaluate uncertainty at this pose
    result_i = evaluate_pose_uncertainty(theta_i, alpha, a, d, Lambda_theta);

    % Store main results
    XYZ(idx,:) = result_i.p;
    reach_vals(idx) = norm(result_i.p);

    U95_vals(idx) = result_i.U95;
    u_max_vals(idx) = result_i.sigma_axes(1);

    sigma1_vals(idx) = result_i.sigma_axes(1);
    sigma2_vals(idx) = result_i.sigma_axes(2);
    sigma3_vals(idx) = result_i.sigma_axes(3);

    sigmaR_vals(idx) = result_i.sigmaR;
    sigmaT_vals(idx) = result_i.sigmaT;
    sigmaZ_vals(idx) = result_i.sigmaZ;

    anisotropy_vals(idx) = result_i.anisotropy;

    if STORE_ALL_ANGLES
        theta_deg_all(idx,:) = theta_deg_i;
    end

    % Track worst and best poses without needing to store every angle
    if result_i.U95 > maxU95
        maxU95 = result_i.U95;
        worst_pose_deg = theta_deg_i;
        worst_position = result_i.p;
    end

    if result_i.U95 < minU95
        minU95 = result_i.U95;
        best_pose_deg = theta_deg_i;
        best_position = result_i.p;
    end
end

```

(b)

Figure A.4: Evaluation of poses in the sampled working space

From here, a distribution of the position uncertainty is plotted to show where uncertainty is the weakest (highlighted in a deep blue) and where it is strongest (highlighted in a bright yellow)¹.

¹See Figure 2.2 under Section 2.3.

Appendix B

Wiring Diagram

List of references

- [1] L. Engineering, "The advantages of using a portable cmm arm in metrology," *Leussink Engineering*, 2024.
- [2] S. Ibaraki and R. Saito, "Novel kinematic model of articulated arm coordinate measuring machine with angular position measurement errors of rotary axes," *CIRP Annals*, vol. 72, no. 1, pp. 449–452, 2023.
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